

A Marvellous Nonsense

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Lewis Carroll describes Snarks to be these elusive creatures who you should fear, and keep an eye open for, especially if it is the *Boojum*— the most dangerous of the Snarks, a threat to your existence.

This existential dread induced by a Snark becomes clearer in Martin Gardner's *The Annotated Snark*, where he equates the Snark with humanity's endless search for meaning— which is agonizing and elusive in the same way.

*"The first is the taste,
Which is meagre and hollow, but crisp:
Like a coat that is rather too tight in the waist,
With a flavour of Will-o'-the-wisp."*

In 1976, Gardner bridged his love for Lewis Carroll and recreational mathematics by popularizing the term Snark in an edition of *Mathematical Games*, namely: Snarks, Boojums and other conjectures related to the four-color-map theorem.

*"Its habit of getting up late you'll agree
That it carries too far, when I say
That it frequently breakfasts at five-o'clock tea,
And dines on the following day."*

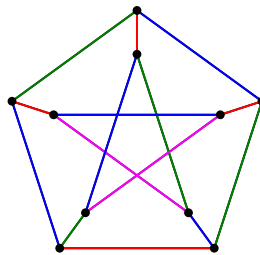
Just like the ship's crew in the poem, the hunt for the Snark is just as important for graph theorists. It is deeply challenging, and whenever a Snark is found, it is a significant result for the field.

*"The third is its slowness in taking a jest.
Should you happen to venture on one,
It will sigh like a thing that is deeply distressed:
And it always looks grave at a pun."*

What is a Snark for a graph theorist after all? In order to understand that, you must leave your *Cartesian Coordinates* behind; with nothing but dots and lines, arrangements emerge that will blow your mind.

*"The fourth is its fondness for bathing-machines,
Which it constantly carries about,
And believes that they add to the beauty of scenes—
A sentiment open to doubt"*

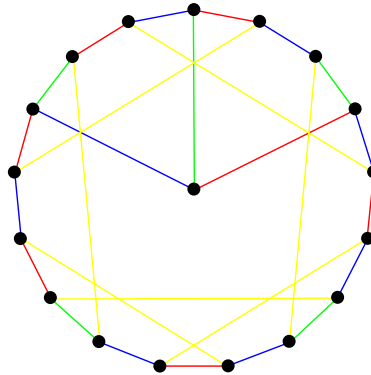
A Snark is an object in graph theory, which has a specific structure where every vertex (node) has three edges (lines) coming out of it, each connected to a distinct vertex. And to colour its edges, it requires at least 4 colours such that no two adjacent edges share the same colour. We can see these properties in the Petersen graph: The Petersen graph was the first Snark discovered by Julius



Petersen in 1898, and it remained so for the next 50 years. Afterwards, when more Snarks were being discovered, Gardner proposed to name them Snarks rather than *nontrivial uncolorable trivalent graphs*.

*The fifth is ambition. It next will be right
To describe each particular batch:
Distinguishing those that have feathers, and bite,
From those that have whiskers, and scratch.*

The next Snarks are the Blanuša Snarks discovered by a Croatian mathematician, Danilo Blanuša, in 1946. One of them uses 18 vertices and the other one uses 26 vertices. The Blanuša Snark on 18 vertices is the 2nd smallest Snark after the Petersen graph. This Snark was first popularised by the Croatian math-



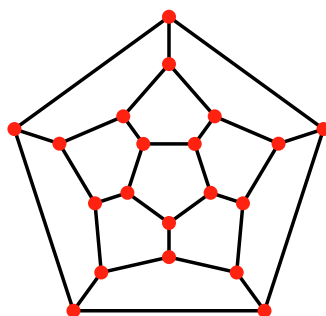
ematics society in 1996, then later it appeared on a Croatian stamp around the turn of the millennium, as the year 2000 was declared *World Mathematical Year* at an *IMU* conference (Rio de Janeiro, 1992); other countries also participated with their mathematical stamps. We don't know what is the *Boojum* of graph



theory, but we know what it has to look like— the hunt goes on...The *Boojum* has all the properties of a Snark with one additional property— when you draw its edges they should not cross or overlap; not even a pair of edges! Such graphs are called *planar graphs*. The *Boojum* is a planar Snark.

*"For, although common Snarks do no manner of harm,
Yet, I feel it my duty to say,
Some are Boojums— " The Bellman broke off in
alarm,"*

If a *Boojum* exists it must look something like the following graph. It does not have any crossing edges, all the vertices are of degree 3 but, its edges can be coloured with three colours, hence it is not a Snark or a *Boojum*. I invite you to try colouring its edges.



*"Come, listen, my men, while I tell you again
The five unmistakable marks*

*By which you may know, wheresoever you go,
The warranted genuine Snarks"*

But where is the nonsense, or what is nonsense? Apart from the stylistic choice of *nonsense*, everything about Snarks seems sensible and reasonable. Doesn't it?

Carroll wrote this poem during the Victorian era when literature was supposed to teach moral values to kids, tell them how to behave and inculcate discipline. Carroll's free-spirited writing broke those traditions and paved the way for more imaginative things. Before graph theory was a proper discipline, the *Bourbaki* style of writing, with dense rigour and formality was common. There was a deliberate exclusion of diagrams and graph theory was considered merely "parlour games" or "recreations".

Gardner himself sat at exactly this intersection, a man who wrote about mathematics in newspapers, and celebrated the communal joy of mathematics. Perhaps we can say Gardner rescued the Snark, refusing to let graph theorists hide behind impenetrable terminology.

Am I the nonsense for drawing this parallel? Will I be exonerated of my *informalities*?

*The weather of Prague is about to make everything melt,
when you plan to cool down
in your chill bath, keep your eyes open,
you may find a Snark.*